

Empowering
Development
through Data-
driven Innovation
and Information
Technology
Solutions

CWIS Indicators in IMIS

Integrated Municipal Information System (IMIS)

Innovative Solution Pvt. Ltd (ISPL)

CWIS Indicators in IMIS

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1. CWIS INDICATORS IN IMIS

There are 22 quantitative CWIS indicators that can be generated from Base IMIS for a given year.

SN	Code	Indicator	Outcome
1	EQ-1	Ratio of LIC access to total population access	equity
2	SF-1a	Percentage of population with access to safe, private, individual toilets/latrines	safety
3	SF-1b	Percentage of on-site sanitation that have been desludged	safety
4	SF-1c	Percentage of collected FS disposed at a treatment plant or at designated disposal site	safety
5	SF-1d	FS treatment capacity as a percentage of total FS generated from NSS connections (excluding safely disposed in situ)	safety
6	SF-1e	FS treatment capacity as a percentage of total FS collected from NSS connections	safety
7	SF-1f	Wastewater treatment capacity as a percentage of total wastewater generated from sewered connections and greywater generated from non-sewered connections	safety
8	SF-1g	Effectiveness of FS/WW treatment in meeting prescribed standards for effluent discharge and biosolids disposal	safety
9	SF-2a	Percentage LIC population with access to safe individual toilets	safety
10	SF-2b	Percentage of LIC, NSS, IHHLs that have been desludged	safety
11	SF-2c	Percentage of collected FS (collected from LIC) disposed at treatment plant or designated disposal sites	safety
12	SF-3	Percentage of dependent population (those without access to a private toilet/latrine) with access to safe shared facilities (CT/PT)	safety
13	SF-3b	Percentage of CTs that adhere to principles of universal design	safety
14	SF-3c	Percentage of users of CTs that are women	safety
15	SF-3e	Average distance from the house to the closest CT (in meters)	safety
16	SF-4a	Percentage of PTs where FS and WW generated is safely transported to TP or safely disposed in situ	safety
17	SF-4b	Percentage of PTs that adhere to principles of universal design	safety
18	SF-4d	Percentage of users of PTs that are women	safety
19	SF-5	Percentage of educational institutions where FS/WW generated is safely transported to TP or safely disposed in situ	safety

20	SF-6	Percentage of healthcare facilities where FS/WW generated is safely transported to TP or safely disposed in situ	safety
21	SF-7	Percentage of desludging services completed mechanically or semi-mechanically (by a gulper)	safety
22	SF-9	Percentage of tests which are in compliance with water quality standards for fecal coliform	safety

2. METHODOLOGY OF COMPUTING CWIS INDICATORS

Structure of data points of Quantitative Indicators				
Information	Methodology mentioned in Survey Tool Kits (ref: Athena)			IMIS
	Household survey	Toilet inspection survey	Desludging services survey	Complete information
Household information	Estimation based on sample household			All households of the city - BIMS
Facilities information	Estimation based on sample household	Estimation based on sample toilet		All households & toilets of the city - FSMIS
Work environment of desludging services			Estimation based on sample desludging service	All desludging services of the city - FSMIS
Sanitation access of household	Estimation based on sample household	Estimation based on sample toilet		All households' sanitation access - BIMS
Collection practices			Estimation based on sample desludging service	All households' collection practices - FSMIS
Containment & collection	Estimation based on sample household	Estimation based on sample toilet		All households' containment and collection status - FSMIS
Disposal			Estimation based on sample desludging service	All households' volume of emptying sludge and disposed at treatment plant or designated site - - FSMIS

3. DEFINITION OF SAFETY IN CWIS

Most of the quantitative CWIS indicators are based on definition of “Safe Individual Toilets” and “Safely Managed Sanitation System”.

3.1 Safely Managed Sanitation System

id	Sanitation System	Outlet connection	Safe
1	Sewer Network	Connected to Treatment Plant	Yes
2	Drain Network	Connected to either Treatment Plant or drain is covered and lined	Yes
3	Septic Tank	Septic Tank connected to Sewer Network with Treatment Plant	Yes
		Septic Tank connected to Drain Network with either Treatment Plant or drain is covered and lined	Yes
		Septic Tank connected to Soak Pit	Yes
		Septic Tank connected to Water Body	No
		Septic Tank connected to Open Ground	No
		Septic Tank without Outlet Connection	No
		Septic Tank with Unknown Outlet Connection	No
4	Pit/ Holding Tank	Double Pit	Yes
		Lined Pit connected to a Soak Pit	Yes
		Lined Pit connected to Water Body	No
		Lined Pit connected to Open Ground	No
		Lined Pit connected to Sewer Network with Treatment Plant	Yes
		Lined Pit connected to Drain Network with either Treatment Plant or drain is covered and lined	Yes
		Lined Pit without Outlet	No
		Lined Pit with Unknown Outlet Connection	No
5	Onsite Treatment (e.g., Anaerobic Digester/ Biogas, DEWATS)		Yes
6	Composting Toilet (e.g., Ecosan, UDDT, etc.)		Yes
7	Water Body		No
8	Open Ground		No
9	Open Defecation		No

3.2 Logical Structure for Safely Managed Sanitation System

A building is said to have safely managed sanitation system (BWSMSS) if any of following conditions is satisfied

Conditions	C-1	sanitation_system_id= Composting toilet				
	C-2	sanitation_system_id = Onsite Treatment (e.g., Anaerobic Digester/ Biogas, DEWATS)				
	C-3	sanitation_system_id = sewer network	sewer code exist	tp_id exist		
	C-4	sanitation_system_id = drain network	drain code exist	tp_id exist		
	C-5			cover_type = closed & surface_type - lined		
	C-6	sanitation_system_id = Pit/holding	type = double pit			
	C-7		containment_id exist	type = lined pit connected to a soak pit		
	C-8			type = lined pit connected to sewer network	sewer code exist	tp_id exist
	C-9			type = lined pit connected to drain network	drain_code exist	tp_id exist
	C-10					cover_type = closed & surface_type – lined
	C-11	sanitation_system_id = Septic tank Or sanitation_system_id = Shared Septic tank	containment_id exist	type = septic_tank connected to a soak pit		
	C-12			type = septic_tank connected to sewer network	sewer code exist	tp_id exist
	C-13			type = septic_tank connected to drain network	drain_code exist	tp_id exist
	C-14					cover_type = closed & surface_type - lined

3.3 Data points in IMIS for safely managed sanitation system

Data points in IMIS	Corresponding Table in IMIS
toilet_status	buildings
sanitation_system_id	buildings
sewer_code	buildings
treatment_plant_id	sewers
drain_code	buildings
cover_type	drains
surface_type	drains
containment_id	buildings
type	containments
treatment_plant_id	drains

4. CWIS INDICATORS WITH IMIS DEFINITION

Each CWIS indicator is generated from IMIS system. The computation procedure of an equity indicator 'Ratio of LIC access to total population access' with IMIS definition and logical structure is provided below as a sample.

EQ-1 Ratio of LIC access to total population access

EQ-1: Equity of Access to Improved Sanitation

Definition: Ratio of LIC access to total population access.

Calculation:

- The numerator of this ratio is the percentage of LIC population with access to 'safe' individual toilets.
- The denominator of this ratio is the percentage of the total population with access to 'safe' individual toilets.
- Formula for the Indicator = $(a/b) / (c/d)$

Where,

a = Population using safe individual/private toilets in LIC

b = Total population in LIC

c = Population using safe individual/private toilets in city

d = Total population in city

S N	CWIS Outcome	Indicator_Code	Indicator Name	Unit	Definition	Calculation	Formula	Description in IMIS
1	Equity	EQ-1	Equity of Access to Improved Sanitation	ratio	Ratio of LIC access to total population access	The numerator of this ratio is the percentage of LIC population with access to 'safe' individual toilets. The denominator of this ratio is the percentage of the total population with access to 'safe' individual toilets.	$(a/b*100) / (c/d*100)$	a= number of persons using private toilet in LICs
								b = Total population in LICs
								c= number of persons using private toilet
								d= total population

Data points	Formula	Description in IMIS	Data points in IMIS	Corresponding Table in IMIS		Logical Statement in IMIS	operation	
a= LIC population with access to safe individual toilets	$(a/b*100)$ $(c/d*100)$	a= population of households having private toilet in LIC	population_served	buildings		a = sum{population_with_private_toilet} with following conditions		
b= Total LIC population		b= population of all households in LIC	toilet_status	buildings	1	toilet_status = yes	and	
c= Population with access to safe individual toilets		c= population of households having private toilet in city	population_with_private_toilet	buildings	2	low_income_hh=yes	or	lic_id exists
d= Total population		d= population of all households in city	low_income_hh	buildings		b = sum {population_served} with following condition		
					3	low_income_hh=yes	or	lic_id exists
						a = sum{population_with_private_toilet} with following conditions		
					4	toilet_status = yes		
						d = sum {population_served}		